

Program Counter

CLOCK – Design Source File

```
-- Group 17
-- Senevirathne S.M.P.U.
--
-- Create Date: 04/14/2024 01:36:22 AM
-- Design Name: Program_Counter
-- Module Name: CLOCK - Behavioral
-- Project Name: Nanoprocessor
-- Target Devices: Basys3 Board
```

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity CLOCK is
    Port ( Clk_in : in STD_LOGIC; --clock input signal with frequency f
          Clk_out : out STD_LOGIC); --modified clock output signal with frequency f/n
end CLOCK;
```

architecture Behavioral of CLOCK is

```

--signal count and Clk_status

signal count : integer := 1;
signal Clk_status : STD_LOGIC := '0';

begin

process (Clk_in) begin
    if (rising_edge (Clk_in)) then
        count <= Count +1;

        if (count =5) then --change 5 to a desired n value. Then the Clk_out frequency will be
1/n th of Clk_in frequency.

            Clk_status <= not Clk_status;

            Clk_out <= Clk_status;

            count <=1;

        end if;

    end if;
end process;

end Behavioral;

```

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-- Module Name: CLOCK - Behavioral
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library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity Program_Counter is

Port (CLK : in STD_LOGIC; --input CLOCK signal

RESET : in STD_LOGIC; --Reset signal

M_Mux : in STD_LOGIC_VECTOR (2 downto 0); --Memory select input value from Mux

M_Sel : out STD_LOGIC_VECTOR (2 downto 0)); --Memory select output value to ROM

end Program_Counter;

architecture Behavioral of Program_Counter is

component CLOCK

port(

Clk_in : in STD_LOGIC;

Clk_out : out STD_LOGIC);

end component;

--signal Clk_out

SIGNAL Clk_out : std_logic;

begin

```
--mapping CLOCK
```

```
CLOCK_PC : CLOCK
```

```
port map (
```

```
    Clk_in => CLK,
```

```
    Clk_out => Clk_out
```

```
);
```

```
--Defining Memory Select output values
```

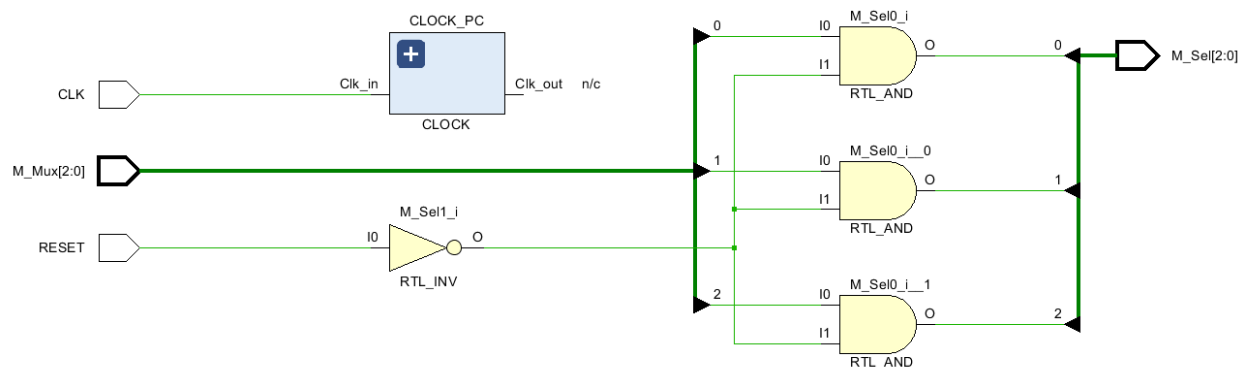
```
M_Sel(2) <= M_Mux(2) AND NOT(RESET);
```

```
M_Sel(1) <= M_Mux(1) AND NOT(RESET);
```

```
M_Sel(0) <= M_Mux(0) AND NOT(RESET);
```

```
end Behavioral;
```

Program Counter – Elaborated Design



Program Counter – Simulation Source file

```
-- Group 17
-- Senevirathne S.M.P.U.
--
-- Create Date: 04/14/2024 01:36:22 AM
-- Design Name: Program_Counter
-- Module Name: Program_Counter_TB - Behavioral
-- Project Name: Nanoprocessor
-- Target Devices: Basys3 Board
```

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity Program_Counter_TB is
end Program_Counter_TB;
```

```
architecture Behavioral of Program_Counter_TB is
```

```
    constant CLOCK_PERIOD : time := 10 ns; -- Define clock period (10 ns)
```

```
    -- Signals for clock module
```

```
    signal clk_tb   : std_logic := '0';
```

```
    -- Signals for Program_Counter module
```

```
    signal reset_tb : std_logic := '0';
```

```
    signal m_mux_tb : std_logic_vector(2 downto 0) := "000";
```

```

signal m_sel_tb : std_logic_vector(2 downto 0);

-- Component instantiation
component CLOCK
    Port ( Clk_in : in STD_LOGIC;
           Clk_out : out STD_LOGIC);
end component;

component Program_Counter
    Port ( CLK : in STD_LOGIC;
           RESET : in STD_LOGIC;
           M_Mux : in STD_LOGIC_VECTOR (2 downto 0);
           M_Sel : out STD_LOGIC_VECTOR (2 downto 0));
end component;

begin

-- Instantiate CLOCK module
clk_gen : CLOCK
    port map (
        Clk_in => clk_tb,
        Clk_out => open -- We don't need the output of the clock module in the testbench
    );

-- Instantiate Program_Counter module
pc : Program_Counter
    port map (
        CLK => clk_tb,
        RESET => reset_tb,

```

```

    M_Mux => m_mux_tb,
    M_Sel => m_sel_tb
);

-- Clock process to generate clock signal
clk_process : process
begin
    while now < 1000 ns loop -- Simulate for 1000 ns
        clk_tb <= '0';
        wait for CLOCK_PERIOD / 2;
        clk_tb <= '1';
        wait for CLOCK_PERIOD / 2;
    end loop;
    wait; -- End simulation
end process;

-- Stimulus process to apply inputs
stimulus : process
begin
    reset_tb <= '1'; -- Assert reset
    m_mux_tb <= "000"; -- Set initial M_Mux value

    wait for 20 ns;

    reset_tb <= '0'; -- De-assert reset

    -- Apply some inputs to M_Mux
    m_mux_tb <= "001";
    wait for 20 ns;

```

```

m_mux_tb <= "010";

wait for 20 ns;

m_mux_tb <= "011";

wait for 20 ns;

```

-- You can continue applying more inputs here

```

wait;

end process;

end Behavioral;

```

Program Counter – Timing Diagram

